Beyond Random-Walk Metropolis

Gibbs sampling, Langevin dynamics, MALA and Hamiltonian Monte Carlo

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Where we are

- Last time: a Markov chain with stationary π, irreducible and aperiodic, gives $\frac{1}{n}\sum_k f(X_k) \to \mathbb{E}_\pi f$ (ergodic theorem).
- Metropolis–Hastings builds such a chain for any $\pi \propto w$: propose $y \sim q(\cdot \mid x)$, accept with $\min\{1, \frac{w(y)q(x|y)}{w(x)q(y|x)}\}$. Detailed balance $\Rightarrow$ stationarity.
- **Random-walk Metropolis** proposes *blindly* (only evaluates w): a random walk needing $O(d)$ steps per independent sample.

Today: use more structure of π.

method	uses	idea
Gibbs	conditional distributions $\pi(x_i \mid x_{-i})$	update one coordinate exactly
Hamiltonian MC	gradient and momentum	long, directed moves
Langevin / MALA	gradient $\nabla \log \pi$	HMC with one step; discretized diffusion

Examples: correlated 2D Gaussian; 1D bimodal target $\pi \propto e^{-U}$, $U(x) = x^4/4 - x^2 - 0.5x$ (last time).

Gibbs sampling

Gibbs sampling

Write $x = (x_1, \ldots, x_d)$ and x_{-i} = all coordinates except x_i. The **full conditional** $\pi(x_i \mid x_{-i})$ is often a standard distribution even when π is not.

Algorithm 1 Gibbs sampler (systematic scan)

for $k = 0, 1, 2, \ldots$ **do**
  for $i = 1, \ldots, d$ **do**
    draw $x_i \sim \pi(x_i \mid x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)$ ▷ using the newest values

- *Random scan*: pick i uniformly at random instead of in order.
- *Blocked Gibbs*: update groups of coordinates jointly from their joint conditional.
- History: Geman & Geman (1984), image restoration; Gelfand & Smith (1990) brought it to Bayesian statistics.

Why Gibbs is correct

Proposition: a Gibbs update is Metropolis–Hastings with acceptance probability 1

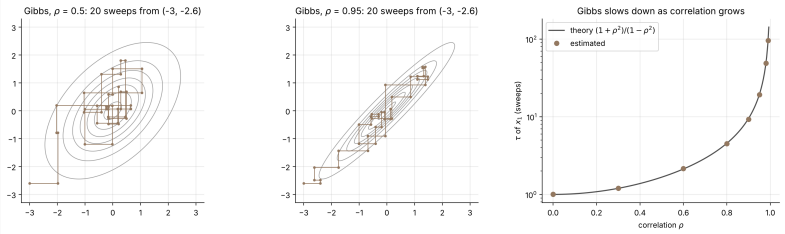
Update coordinate i with the proposal $q(y \mid x) = \pi(y_i \mid x_{-i})$, $y_{-i} = x_{-i}$. Using $\pi(x) = \pi(x_i \mid x_{-i})\,\pi(x_{-i})$:

$$\frac{\pi(y)\,q(x \mid y)}{\pi(x)\,q(y \mid x)} = \frac{\pi(y_i \mid x_{-i})\,\pi(x_{-i}) \cdot \pi(x_i \mid x_{-i})}{\pi(x_i \mid x_{-i})\,\pi(x_{-i}) \cdot \pi(y_i \mid x_{-i})} = 1.$$

- So each single-coordinate update satisfies detailed balance and leaves π invariant.
- A sweep is a *composition* of π-invariant kernels, hence π-invariant. (Systematic scan is invariant but not reversible; random scan is reversible.)
- **Ergodicity** needs more: e.g. if $\pi > 0$ on a connected open set with positive conditionals, Gibbs is irreducible and aperiodic.
- **Failure:** π uniform on two squares $[0,1]^2 \cup [2,3]^2$: axis-aligned moves can never jump between them.

Example: correlated Gaussian

$\pi = \mathcal{N}\big(0, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\big)$: $\quad x_1 \mid x_2 \sim \mathcal{N}(\rho x_2, 1-\rho^2), \quad x_2 \mid x_1 \sim \mathcal{N}(\rho x_1, 1-\rho^2).$



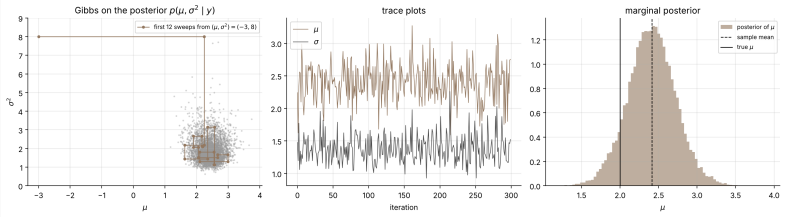
- Moves are axis-aligned; with strong correlation each step is short compared with the long axis.
- x_1 along the sweeps is an AR(1) chain with coefficient ρ^2, so $\tau = \frac{1+\rho^2}{1-\rho^2}$: $\rho = 0.99$ gives $\tau \approx 100$.
- Remedy: rotate coordinates or update correlated coordinates as a block.

A Bayesian example: normal model with unknown mean and variance

Data $y_1, \dots, y_n \sim \mathcal{N}(\mu, \sigma^2)$; priors $\mu \sim \mathcal{N}(m_0, s_0^2)$, $\sigma^2 \sim \mathrm{InvGamma}(a_0, b_0)$. The joint posterior is not standard, but the conditionals are:

$$\mu \mid \sigma^2, y \sim \mathcal{N}\Big(v\big(\tfrac{m_0}{s_0^2} + \tfrac{\sum_i y_i}{\sigma^2}\big),\, v\Big), \quad v = \Big(\tfrac{1}{s_0^2} + \tfrac{n}{\sigma^2}\Big)^{-1};$$

$$\sigma^2 \mid \mu, y \sim \mathrm{InvGamma}\Big(a_0 + \tfrac{n}{2},\; b_0 + \tfrac{1}{2}\textstyle\sum_i (y_i - \mu)^2\Big).$$



$n = 20$ observations: the chain reaches the posterior in two sweeps, and $\tau \approx 1$ for both μ and σ^2: almost independent draws.

When to use Gibbs

- **Conditionally conjugate models**: hierarchical models (group means, variance components), Bayesian linear regression, mixture models (cluster labels | parameters, parameters | labels), latent-variable models, LDA.
- **Data augmentation**: add latent variables that make conditionals simple (e.g. probit or Pólya–Gamma augmentation for logistic regression).
- **Metropolis-within-Gibbs**: if one conditional is not standard, update that coordinate with an MH step targeting $\pi(x_i \mid x_{-i})$.

Limitations

- Slow under strong posterior correlation (axis-aligned moves). Reparametrize or block.
- Needs tractable conditionals: rarely the case for neural-network weights.
- For general differentiable posteriors, gradients are the more useful structure. $\Rightarrow$ Langevin and HMC.

Hamiltonian Monte Carlo

Hamiltonian dynamics

The kinetic energy $T = \frac{|v|^2}{2}$. (In general $\frac{1}{2} p^\top M^{-1} p$, we assume the unity mass, then momentum is equal to the velocity to avoid confusion with probability density.)

The Newton equation in the potential field U (which depends on x only) is $\ddot{x} = -\frac{\partial U}{\partial x}$ and $v = \dot{x}$. Consider their sum (Hamiltonian):

$$H(x, v) = T(v) + U(x).$$

$$\dot{x} = \frac{\partial H}{\partial v} = \frac{\partial T}{\partial v} = v, \qquad \dot{v} = -\frac{\partial H}{\partial x} = -\frac{\partial U}{\partial x} = \ddot{x}.$$

Three properties (exact flow Φ_t):

1. **Energy conservation**: $\frac{\mathrm{d}}{\mathrm{d}t} H = \nabla U \cdot v + v \cdot (-\nabla U) = 0$, so e^{-H} is constant along trajectories.
2. **Volume preservation** (Liouville): the vector field has zero divergence, $\partial_x \cdot v + \partial_v \cdot (-\nabla U) = 0$.
3. **Time reversibility**: flip the momentum, run for time t, flip again: you return to the start.

Augmented target

Our goal is to sample from unnormalized probability $p(x) = \frac{w(x)}{Z}$. Because we are interested in the maximum of probability, introdue the potential energy as

$$U(x) = -\log p(x) = -\log w(x) + \log Z.$$

Introduce an auxiliary momentum $v \in \mathbb{R}^d$ and the **Hamiltonian** (total energy)

$$H(x, v) = U(x) + \frac{|v|^2}{2}, \qquad \pi(x, v) \propto e^{-H(x,v)} = e^{-U(x)} \cdot e^{-\frac{1}{2}|v|^2}.$$

- x and v are independent, $v \sim \mathcal{N}(0, I)$, and the x-marginal is our target $\pi(x)$.
- So: sample (x, v) from $\pi(x, v)$ and throw v away.

Consequence: Φ_t maps $\pi(x, v)$ to itself (1+2), but it stays on one energy level set, so it cannot sample π alone. Fix: **resample the momentum** between trajectories. This changes the energy level.

Stationarity proof (exact Hamiltonian flow)

One step: draw $v \sim \mathcal{N}(0, I)$, set $(x', v') = \Phi(x, v)$ ($\Phi = \Phi_T$, exact flow for time T), keep x':

$$q(x', v' \mid x, v) = \delta\big((x', v') - \Phi(x, v)\big), \qquad q(x' \mid x) = \int \Big[\int q(x', v' \mid x, v) \underbrace{p(v \mid x)}_{= p(v)} \, \mathrm{d}v \Big] \, \mathrm{d}v'.$$

Claim: $\int p(x)\, q(x' \mid x)\, \mathrm{d}x = p(x')$. **Proof.** Since $p(x)p(v) = \pi(x, v)$,

$$\int p(x)\, q(x' \mid x)\, \mathrm{d}x = \int \Big[\iint \pi(x, v)\, \delta\big((x', v') - \Phi(x, v)\big)\, \mathrm{d}x\, \mathrm{d}v \Big] \, \mathrm{d}v'.$$

Substitute $(y, u) = \Phi(x, v)$. By volume preservation (2), $\mathrm{d}x\, \mathrm{d}v = \mathrm{d}y\, \mathrm{d}u$; by energy conservation (1), $\pi\big(\Phi^{-1}(y, u)\big) = \pi(y, u)$. The inner integral becomes

$$\iint \pi\big(\Phi^{-1}(y, u)\big)\, \delta\big((x', v') - (y, u)\big)\, \mathrm{d}y\, \mathrm{d}u = \pi(x', v') \qquad \text{(the flow maps } \pi \text{ to itself).}$$

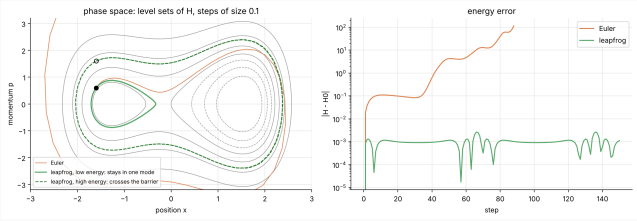
Therefore $\int p(x)\, q(x' \mid x)\, \mathrm{d}x = \int \pi(x', v')\, \mathrm{d}v' = p(x') \int p(v')\, \mathrm{d}v' = p(x').$ ∎

Only (1) and (2) were used: exact flow preserves $\pi(x, v)$, and drawing a fresh v is an exact Gibbs step. Z never appears.

Leapfrog integration

The exact flow is not computable; use the **leapfrog** (Störmer–Verlet) scheme with step ϵ:

$$v_{t+\frac{1}{2}} = v_t - \frac{\epsilon}{2}\frac{\partial H}{\partial x} = v_t + \frac{\epsilon}{2}\frac{\partial \log p(x_t)}{\partial x}, \qquad x_{t+1} = x_t + \epsilon\, v_{t+\frac{1}{2}}, \qquad v_{t+1} = v_{t+\frac{1}{2}} + \frac{\epsilon}{2}\frac{\partial \log p(x_{t+1})}{\partial x}.$$



- Each sub-step is a shear: **volume-preserving**. The scheme is symmetric: **reversible**.
- It does not conserve H exactly, but the error stays bounded, $O(\epsilon^2)$ (symplectic). Euler gains energy and spirals out.
- Left: a low-energy orbit stays in one mode; a high-energy orbit crosses the barrier.

Leapfrog: where reversibility (3) is needed

The leapfrog map Ψ (L steps) preserves volume but *not* H, so the proof above breaks. Repair it with a Metropolis step.

- Let $F(x, v) = (x, -v)$ (momentum flip) and $S = F \circ \Psi$. By reversibility (3) of leapfrog, $S \circ S = \mathrm{id}$: S is an **involution**. Also $|\det \nabla S| = 1$.
- Propose $z' = S(z)$, $z = (x, v)$, and accept with $\alpha(z) = \min\{1, \pi(Sz)/\pi(z)\} = \min\{1, e^{H(z) - H(Sz)}\}$.
- Detailed balance: the flow of probability from z to Sz is

$$\pi(z)\,\alpha(z) = \min\{\pi(z), \pi(Sz)\},$$

  symmetric under $z \leftrightarrow Sz$, and S maps Sz back to z. Since S preserves volume, no Jacobian appears. Hence $\pi(x, v)$ is stationary.
- F does not change H and is irrelevant afterwards, because v is resampled at the next step.
- With the exact flow, $H(Sz) = H(z)$, so $\alpha = 1$: the previous slide is the special case of perfect integration.

Deterministic proposals: the Jacobian replaces the proposal ratio

Proposition (Green 1995)

Let S be an involution ($S \circ S = \mathrm{id}$), $J(z) = |\det \nabla S(z)|$. Propose $z' = S(z)$, accept with $\alpha(z) = \min\left\{1, \dfrac{\pi(Sz)}{\pi(z)} J(z)\right\}$. Then detailed balance holds.

Heuristic. $q(z' \mid z) = \delta(z' - Sz) = J(z')\,\delta(z - Sz')$ and $q(z \mid z') = \delta(z - Sz')$, so $\dfrac{q(z \mid z')}{q(z' \mid z)} = \dfrac{1}{J(z')} = J(z)$ (involution: $J(Sz)J(z) = 1$).

Proof. Rejections are symmetric. For acceptances, substitute $z = S(w)$, $\mathrm{d}z = J(w)\,\mathrm{d}w$:

$$\int \pi(z)\alpha(z)\,1_A(z)1_B(Sz)\,\mathrm{d}z = \int \pi(Sw)\alpha(Sw)J(w)\,1_A(Sw)1_B(w)\,\mathrm{d}w,$$

and $\pi(Sw)J(w)\alpha(Sw) = \min\{\pi(Sw)J(w), \pi(w)\} = \pi(w)\alpha(w)$: symmetric in A, B. ■

HMC: $S = F \circ \Psi$ is an involution by reversibility (3) ($\Psi^{-1} = F\Psi F$), and $J = 1$ by volume preservation, so $\alpha = \min\{1, e^{H(z) - H(Sz)}\}$. Without the flip F: $q(z \mid z') = 0$, every move rejected.

The HMC algorithm

Algorithm 2 Hamiltonian Monte Carlo

1: **for** $k = 0, 1, 2, \ldots$ **do**
2: draw a fresh momentum $v \sim \mathcal{N}(0, I)$
3: $(x', v') \leftarrow L$ leapfrog steps of size ϵ from (x_k, v)
4: accept: $x_{k+1} = x'$ with probability $\min\{1, \exp\big(H(x_k, v) - H(x', v')\big)\}$; otherwise $x_{k+1} = x_k$

- Cost: L gradient evaluations per iteration.
- Exact Hamiltonian flow would give acceptance 1; leapfrog's small energy error keeps acceptance high even for long trajectories.
- The accept/reject step removes the discretization bias exactly (unlike ULA).

Stationarity is not ergodicity

The ergodic theorem also needs **irreducibility** and **aperiodicity**.

- **Irreducibility:** the fresh momentum puts the chain on a new energy level at every step. For smooth U and small enough T, the map $v \mapsto x'$ is a local diffeomorphism near $x' \approx x + Tv$, so x' has a positive density around x; chaining steps, every set of positive probability is reached.
- **Periodicity can happen** for special T. Example: $p = \mathcal{N}(0, 1)$, $U = x^2/2$, so $x(t) = x \cos t + v \sin t$.
  - $T = \pi$: $x' = -x$ for every v. The chain alternates $x, -x, x, \ldots$: stationary but not ergodic.
  - $T = 2\pi$: $x' = x$, the chain never moves.
- **Remedy in practice:** randomize the trajectory length (T or L drawn at random each iteration), or choose it adaptively (NUTS).

Why HMC is correct

An HMC iteration is a composition of two $\pi(x, v)$-invariant kernels:

1. **Momentum refresh**: draw v from its exact conditional $\pi(v \mid x) = \mathcal{N}(0, I)$. A Gibbs step: invariant.
2. **Metropolis step with a deterministic proposal**: $S =$ (flip momentum) $\circ$ (L leapfrog steps).
   - S is an *involution* ($S \circ S = \mathrm{id}$) by reversibility, and preserves volume ($|\det \nabla S| = 1$).
   - For such a proposal the MH ratio is just $\pi(S(z))/\pi(z) = e^{H(z) - H(S(z))}$: this is the acceptance rule. Detailed balance holds.
   - The final momentum flip does not matter, because v is resampled next.

Composition of invariant kernels is invariant $\Rightarrow \pi(x, v)$ is stationary, so the x-chain samples $p(x)$. Ergodicity comes from the random momentum (and, in practice, a randomly jittered L to avoid periodic orbits).

Tuning HMC

- **Step size** ϵ: too large $\Rightarrow$ energy error, rejections, "divergent" trajectories; target acceptance ≈ 0.65–0.8.
- **Trajectory length** $L\epsilon$: too short $\Rightarrow$ random walk again; too long $\Rightarrow$ the trajectory turns back (wasted gradients). The **No-U-Turn Sampler** (NUTS; Hoffman & Gelman 2014) chooses L adaptively; it is the default in Stan, PyMC, NumPyro.
- **Mass matrix** M: set $M^{-1} \approx$ posterior covariance (estimated during warm-up). Equivalent to rescaling x so the target is round.
- **Limits**: HMC still struggles with multimodality and with very varying curvature (funnels); it needs a differentiable U.

Discrete Langevin as a particular case of HMC

HMC with one leapfrog step

Take $L = 1$ and a fresh momentum $v \sim \mathcal{N}(0, I)$. One leapfrog step from (x, v):

$$v_{\frac{1}{2}} = v + \frac{\epsilon}{2} \nabla \log p(x), \qquad x' = x + \epsilon\, v_{\frac{1}{2}} = x + \frac{\epsilon^2}{2} \nabla \log p(x) + \epsilon\, v.$$

Put $\boxed{h = \epsilon^2/2}$, i.e. $\epsilon = \sqrt{2h}$, and rename $v = \xi$:

$$\boxed{x' = x + h \nabla \log p(x) + \sqrt{2h}\, \xi, \qquad \xi \sim \mathcal{N}(0, I)}$$

This is the **discrete Langevin step**.

- The new position needs only $\nabla \log p$ at the *old* point; v' is needed only for the acceptance test.
- **Unadjusted Langevin algorithm (ULA):** repeat the step, skip the accept/reject.
- **Metropolis-adjusted Langevin (MALA):** keep the HMC accept/reject (Roberts & Tweedie 1996).

Reading the Langevin step

$$x_{k+1} = \underbrace{x_k + h\,\nabla \log p(x_k)}_{\text{gradient ascent on } \log p} + \underbrace{\sqrt{2h}\,\xi_k}_{\text{noise}}.$$

- Without the noise: gradient ascent with learning rate h. It converges to a mode, i.e. the MAP estimate, not to the distribution.
- The noise keeps the chain spread out. Its size is not free: $\sqrt{2h}$ comes from the momentum $v \sim \mathcal{N}(0, I)$ of HMC.
- In Bayesian deep learning: $\log p(\theta \mid \mathcal{D}) = \log p(\theta) + \sum_i \log p(y_i \mid x_i, \theta) + \text{const}$. The gradient is computed by backpropagation, and the constant (Z) is never needed.
- Compared with HMC: Langevin uses the momentum for one step only and then forgets it. Each step goes in a new random direction.

MALA: the acceptance probabilities coincide

As a Metropolis–Hastings proposal, the Langevin step is $q(x' \mid x) = \mathcal{N}\big(x';\, x + h\nabla \log p(x),\, 2hI\big)$. With $v' = v + \frac{\epsilon}{2}\nabla \log p(x) + \frac{\epsilon}{2}\nabla \log p(x')$ (end of the leapfrog step), the residuals are

$$x' - x - h\nabla \log p(x) = \epsilon v, \qquad x - x' - h\nabla \log p(x') = -\epsilon v'.$$

Since $\epsilon^2/(4h) = \frac{1}{2}$: $\log q(x' \mid x) = -\frac{1}{2}|v|^2 + c$ and $\log q(x \mid x') = -\frac{1}{2}|v'|^2 + c$. Hence

$$\frac{p(x')\, q(x \mid x')}{p(x)\, q(x' \mid x)} = \exp\Big(U(x) + \tfrac{1}{2}|v|^2 - U(x') - \tfrac{1}{2}|v'|^2\Big) = e^{H(x,v) - H(x',v')}.$$

MALA and one-step HMC are the same Markov chain. Numerical check on the bimodal target, $h = 0.3$:

	acceptance	mean	variance
MALA, $h = 0.3$	0.717	0.759	1.272
HMC, $L = 1$, $\epsilon = \sqrt{0.6}$	0.718	0.759	1.269
target	–	0.760	1.262

Without the accept/reject step, ULA is biased

Gaussian target $p = \mathcal{N}(0, 1)$, so $\nabla \log p(x) = -x$:

$$x_{k+1} = (1 - h)\, x_k + \sqrt{2h}\, \xi_k.$$

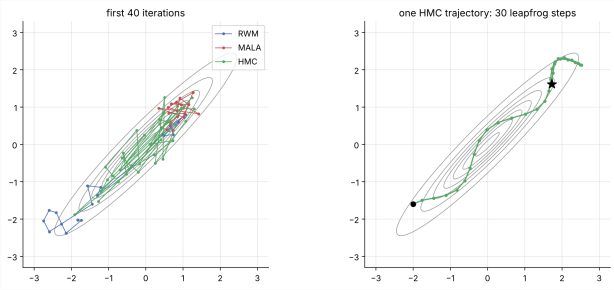
This is the AR(1) chain of last lecture with $a = 1 - h$. Its stationary variance is

$$\sigma_\infty^2 = \frac{2h}{1 - (1 - h)^2} = \frac{1}{1 - h/2} \neq 1.$$

- The AR(1) chain with the *correct* stationary variance needs noise variance $1 - a^2 = 2h - h^2$. ULA uses $2h$: too much noise.
- The bias is $O(h)$. Small h: accurate but slow. Large h: fast but wrong.
- MALA removes the bias for every h: it is MH, so detailed balance holds.

Question: what does ULA approximate as $h \to 0$? $\Rightarrow$ a continuous-time process.

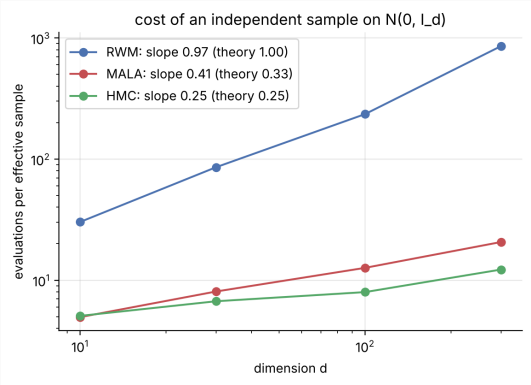
Comparison on a correlated Gaussian ($\rho = 0.95$)



	accept	grads /iter	ESS per 1000 grads
RWM	0.45	1	16
MALA	0.50	1	15
HMC	0.92	10	**119**

- Left: HMC jumps across the whole ellipse every iteration; RWM and MALA crawl.
- Right: one trajectory follows the long axis and turns back (a "U-turn"): the idea behind NUTS.
- In 2D a gradient alone (MALA) does not beat RWM; momentum does.

The price of forgetting: dimension scaling



On $\mathcal{N}(0, I_d)$ with optimally scaled steps, the number of target or gradient evaluations per independent sample grows like

	step size	cost
RWM	$\sigma \propto d^{-1/2}$	$O(d)$
MALA	$h \propto d^{-1/3}$	$O(d^{1/3})$
HMC	$\epsilon \propto d^{-1/4}$, $L\epsilon = O(1)$	$O(d^{1/4})$

(Roberts, Gelman & Gilks 1997; Roberts & Rosenthal 1998; Beskos et al. 2013.) Optimal acceptance: 0.234, 0.574, ≈ 0.65.

For a network with $d = 10^6$ weights: 10^6 vs 100 vs 32.

Summary

method	uses	why π is stationary	cost in d	weak spot
RWM	$w(x)$	MH, detailed balance	$O(d)$	random walk
Gibbs	conditionals	MH with acceptance 1	–	correlations
ULA	∇U	*not exactly*: $O(h)$ bias	–	bias, stiffness
MALA	∇U	MH, detailed balance	$O(d^{1/3})$	random walk
HMC	∇U, momentum	Gibbs on p + MH with involution	$O(d^{1/4})$	tuning ϵ, L; multimodality

1. Gibbs: exact conditional updates; best with conjugate structure, slow under correlation.
2. Langevin = HMC with $L = 1$ ($h = \epsilon^2/2$): gradient ascent on $\log p$ + noise $\sqrt{2h}$; ULA is biased, MALA corrects with MH.
3. HMC: augment with momentum, follow Hamiltonian dynamics with leapfrog, accept with $e^{H-H'}$.
4. ULA is the Euler–Maruyama discretization of the Langevin diffusion $\mathrm{d}X = \nabla \log p \, \mathrm{d}t + \sqrt{2}\, \mathrm{d}W$, whose stationary density is p (Fokker–Planck).

Notebook: `gibbs_langevin_hmc.ipynb`. **Next:** stochastic-gradient MCMC for large data; diffusion models.